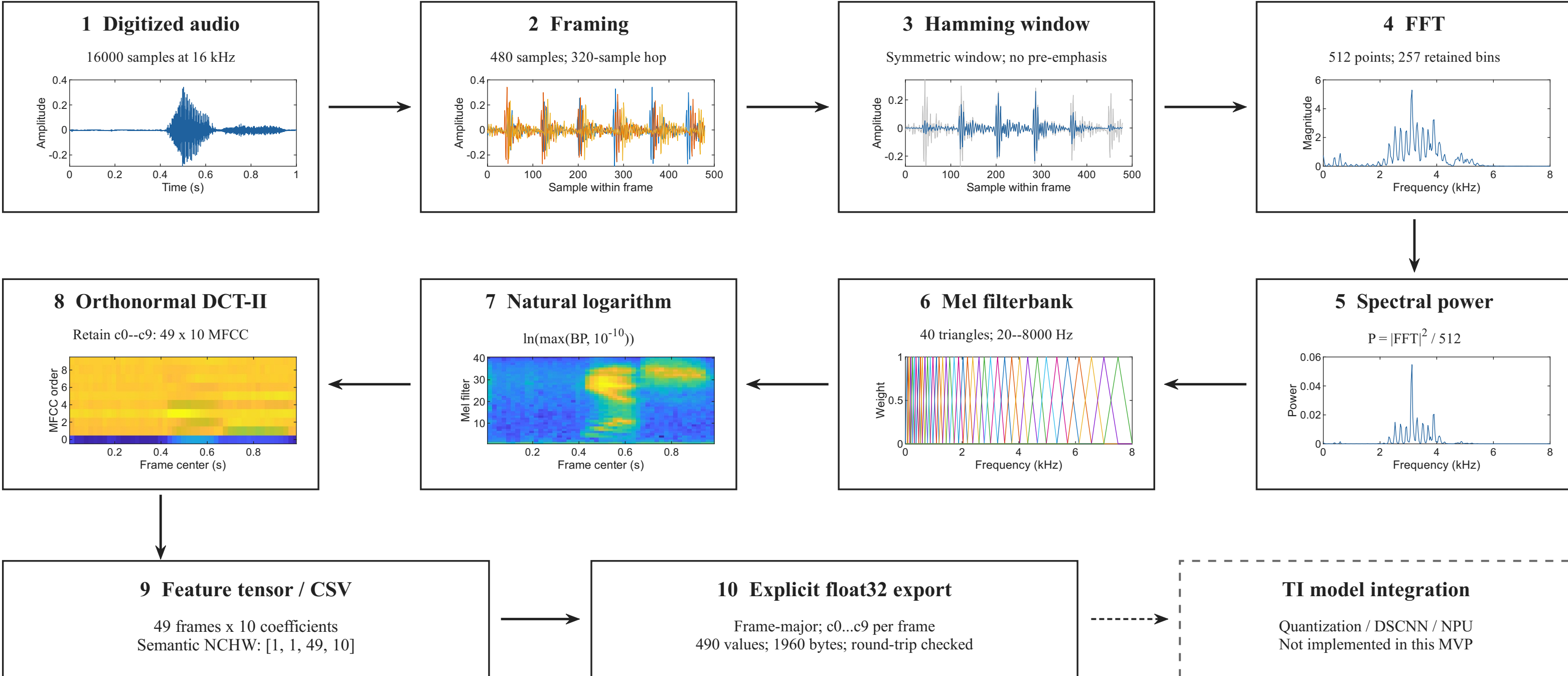


Single-audio MFCC pipeline

Original MATLAB MVP numerical contract | Real recording: yes.wav | feature extraction only



Solid arrows: computed on the same audio. Heatmaps show actual values; see stage figures for color scales.
Baseline-only branch: MFCC mean/std -> training standardization -> centroid distance -> argmin (synthetic model).